

ONON, Mongolia

WMO: 442570

Lat: 48.6161N

Lon: 110.5989E

Elev: 7156

StdP: 11.27

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-25.3	-35.8	0.9	-27.5	-32.1	1.2	-23.4	26.5	1.8	23.0	0.6	0.8	320	0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	86.9	60.2	82.6	59.3	79.0	58.4	64.8	77.9	63.1	75.5	61.5	73.3	9.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	105.8	69.7	59.3	98.6	68.4	57.5	92.3	66.9	34.0	78.2	32.6	74.9	31.2	73.5	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.0	21.7	19.6	DB	-33.8	92.6	4.1	4.5	-36.8	95.9	-39.2	98.5	-41.5	101.0	-44.5	104.2	(4)
(5)			WB	-34.1	68.5	3.9	2.3	-36.9	70.2	-39.2	71.5	-41.4	72.8	-44.3	74.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.0	-6.5	1.8	19.6	36.7	49.5	61.3	66.0	62.1	50.3	33.6	11.2	-3.4	
	DBStd	27.03	9.13	10.13	11.64	8.91	7.88	6.29	4.99	6.20	7.44	8.51	11.98	10.57	
	HDD50	7971	1751	1350	941	405	106	3	0	2	85	509	1164	1655	
	HDD65	12189	2216	1770	1406	848	484	143	44	129	442	973	1614	2120	
	CDD50	1406	0	0	0	7	90	342	496	376	94	1	0	0	
	CDD65	149	0	0	0	0	3	32	75	38	1	0	0	0	
	CDH74	2200	0	0	0	3	147	622	833	533	62	0	0	0	
	CDH80	676	0	0	0	0	41	203	272	150	10	0	0	0	
Wind	WSAvg	6.8	5.2	6.4	7.1	8.9	9.8	7.1	5.7	6.1	7.0	6.5	5.9	5.5	
Precipitation	PrecAvg	12.9	0.1	0.1	0.2	0.4	0.8	2.3	3.8	2.8	1.5	0.5	0.2	0.1	
	PrecMax	20.4	0.2	0.3	0.9	1.2	3.4	5.1	9.2	9.6	4.1	1.8	0.6	0.3	
	PrecMin	6.2	0.0	0.0	0.0	0.0	0.0	0.3	1.3	0.9	0.0	0.0	0.0	0.0	
	PrecStd	3.7	0.1	0.1	0.2	0.3	0.7	1.4	1.9	1.9	1.0	0.4	0.2	0.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	36.4	57.4	71.1	85.7	92.0	93.8	89.3	81.5	66.1	44.4	27.3	
		MCWB	16.0	26.5	38.2	45.9	54.0	60.1	64.0	62.7	56.6	46.0	33.5	21.8	
	2%	DB	15.7	29.9	49.7	64.5	78.2	85.9	87.0	84.2	75.1	59.5	37.7	21.1	
		MCWB	11.7	22.6	34.8	43.0	50.7	58.6	62.3	60.6	53.3	42.1	29.7	17.0	
	5%	DB	11.5	24.4	43.5	59.1	73.0	80.6	81.8	79.9	70.4	53.9	33.3	17.0	
		MCWB	8.4	18.8	31.7	40.4	49.2	56.8	61.3	59.6	52.2	39.2	26.8	13.7	
	10%	DB	7.8	19.4	37.6	53.9	68.0	76.7	78.0	75.9	66.0	48.7	29.1	12.5	
		MCWB	5.3	14.9	28.3	38.4	47.3	55.9	60.6	58.7	50.4	36.8	23.8	9.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	27.1	38.9	47.7	56.1	63.2	68.3	66.1	59.0	46.7	33.7	22.1
			MCDB	21.4	36.1	55.7	66.6	80.1	83.2	81.7	82.2	74.2	64.2	43.3	27.1
2%		WB	11.8	22.7	35.1	44.0	52.8	61.0	65.6	63.9	56.7	42.8	30.2	17.0	
		MCDB	15.7	29.6	48.7	61.9	73.5	76.1	78.5	76.4	69.5	57.8	36.9	20.9	
5%		WB	8.5	18.7	31.9	41.3	50.3	59.5	64.2	62.1	54.6	39.9	27.2	13.5	
		MCDB	11.4	24.2	43.1	57.6	70.0	73.6	76.5	74.1	65.8	52.4	33.0	16.7	
10%		WB	5.4	15.0	28.8	38.9	48.1	57.9	62.7	60.3	52.2	37.4	23.9	9.8	
		MCDB	7.7	19.2	37.1	52.9	65.8	71.5	74.1	71.6	63.6	47.9	29.0	12.3	
Mean Daily Temperature Range	5% DB	MDBR	20.5	25.2	25.5	24.6	26.8	24.4	21.4	23.2	25.4	23.5	20.9	18.5	
		MCDBR	24.4	31.9	32.8	35.2	38.2	34.0	29.5	31.5	34.5	33.4	24.9	22.1	
		MCWBR	21.1	25.2	20.3	18.3	17.0	12.3	10.3	11.9	16.7	18.3	18.0	19.0	
	5% WB	MCDBR	24.2	31.4	31.9	31.5	33.0	25.2	23.0	24.2	27.7	29.6	24.5	22.1	
		MCWBR	21.0	25.1	19.9	16.8	15.7	10.6	9.9	10.3	14.2	16.9	17.9	19.0	
Clear-Sky Solar Irradiance	taub	0.205	0.222	0.251	0.293	0.333	0.341	0.321	0.301	0.270	0.242	0.218	0.205		
	taud	2.448	2.443	2.453	2.404	2.310	2.378	2.480	2.520	2.564	2.576	2.510	2.438		
	Ebn at Noon	292	308	312	305	294	292	296	299	301	297	284	277		
	Edh at Noon	19	24	29	34	39	37	33	30	26	21	17	16		
All-Sky Solar Radiation	RadAvg	586	966	1447	1779	2009	1983	1880	1715	1400	975	617	462		
	RadStd	25	46	62	87	107	121	102	101	101	44	29	37		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	-2.25	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page